

Week 8 Lab Activity

7CS518 — Natural Language Processing

Module Leader: Ioannis Tsioulis

Topic: Sequence-to-Sequence Learning & Attention

This week's lab has two parts. The first reinforces the theoretical content of Unit 8 (encoder–decoder models, the attention mechanism, and translation evaluation). The second gives you a head start on the final assessment by exploring a complete text-classification pipeline on a realistic dataset.

Part 1 — Seq2seq & Attention (theory in practice)

Work through the official PyTorch tutorial below. It implements an encoder–decoder network with Bahdanau attention for English–French translation, and mirrors the concepts covered in the Unit 8 slides: context vector, teacher forcing, attention weights, and attention visualisation. Focus on understanding the code rather than running every cell.

https://docs.pytorch.org/tutorials/intermediate/seq2seq_translation_tutorial.html

Part 2 — Sentiment Analysis Pipeline (assignment preparation)

Explore the notebook below, which performs sentiment analysis on the IMDB 50K movie-review dataset. It covers EDA, text preprocessing, TF-IDF, and a comparison between classical classifiers (Naive Bayes, Logistic Regression, SVM) and deep-learning models (RNN/LSTM, CNN).

https://github.com/ChiragJawale9/Sentiments_Analysis_IMDB_Movie_Reviews
